# Expanding global coverage of climate change support over space and time*

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**Abstract**

Public support is a critical determinant of the success or failure of climate policy. However, data on climate attitudes remain fragmented, inconsistent in format, and difficult to compare across countries and over time. We introduce a globally representative panel dataset capturing public attitudes toward climate change in over 150 countries between 2010 and 2024. The dataset integrates more than 8,000 country-year-item responses from 12 major international and regional surveys, such as the World Risk Poll, Latinobarómetro, Afrobarometer, and the Gallup World Poll. These diverse question formats and response scales are harmonized and standardized. We apply a Bayesian ordinal Item Response Theory (IRT) model with partial pooling to estimate a latent measure of climate support for each country-year, enabling consistent comparisons over time and across regions. The resulting data product significantly improves the temporal continuity and geographic inclusiveness of existing resources, particularly by expanding coverage to many low- and middle-income countries. Organized in an accessible tabular format, this dataset supports longitudinal and cross-national analysis of climate opinion dynamics. It offers a valuable tool for researchers and policymakers engaged in global climate governance, public engagement, and climate communication.

**Keywords:** climate change, public opinion, survey data, Item Response Theory

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# 1 Introduction

Public attitudes toward climate change play a crucial role in shaping national policies and fostering international cooperation (Falkner, 2016). Yet until recently, surprisingly little was known about how these attitudes are distributed globally or how they evolve over time. Over the past decade, large-scale cross-national studies covering more than 100 countries—such as those by Lee et al. (2015), Ballew et al. (2020), and Andre et al. (2024)—have begun to fill this gap. However, important limitations remain. Existing datasets are heavily skewed toward high-income democracies, leaving much of the Global South underrepresented. Moreover, longitudinal data that allow for consistent tracking over time are sparse and mostly limited to a small set of countries. As a result, our understanding of global climate opinion is fragmented, temporally discontinuous, and geographically uneven.

This paper addresses these challenges with three main objectives: (1) to expand geographic coverage to include nearly all countries worldwide; (2) to extend the temporal panel backward beyond OECD countries; and (3) to construct a harmonized country-year panel dataset on climate attitudes covering more than 150 countries from 2010 to 2024.

To achieve this, we compile and harmonize over 8,000 country-year-item observations drawn from major international and regional surveys—including the World Risk Poll, Latinobarómetro, Afrobarometer, and Gallup World Poll. We apply a Bayesian ordinal Item Response Theory (IRT) model with partial pooling to account for differences in question wording, response scales, and survey coverage. This approach enables the estimation of a latent climate support score for each country-year, even when data are missing or survey designs are inconsistent.

Our findings demonstrate that a fully cross-sectional dataset covering more than 165 countries is attainable; a consistent longitudinal panel from 2010 to 2024 can be constructed for nearly 100 countries; and a longer historical series extending back to the early 1990s is feasible for about 40 countries, primarily within the OECD. This paper details the dataset construction, describes the methodology, and discusses its potential to advance research on global climate opinion dynamics across space and time.

# 2 Global data on climate attitudes: coverage, continuity, and concepts

## 2.1 Cross-national data coverage

Early research on public opinions regarding climate change was predominantly country-specific, with a strong emphasis on the United States. However, over the past two decades, there has been significant growth in cross-national studies, seeking to examine how climate change attitudes vary across societies and what national-level factors account for such differences.

Initial cross-national efforts typically included 25 to 50 countries and employed multi-level research designs. For example, Steg and Vlek (2009) investigated psychological determinants of climate change attitudes, though their analysis was limited to European contexts. Similarly, Poortinga et al. (2019) utilized data from the European Social Survey to examine regional differences in climate risk perception and policy support across EU member states.

In recent years, researchers have increasingly leveraged truly global datasets to study climate attitudes. For instance, Lee et al. (2015) analyzed data from 119 countries to identify individual-level predictors of climate change awareness and risk perception. Expanding on this, Knight (2016) used the 2007–2010 Gallup World Poll—then the largest international dataset on climate opinion—for a global analysis of public awareness. More recently, Andre et al. (2024) conducted a landmark survey across 125 countries and nearly 130,000 respondents, setting a new benchmark in global opinion research. Additional large-scale studies, such as Schuldt et al. (2018) and Ballew et al. (2020), have further extended coverage to over 120 countries, examining ideological divides and the role of cultural and economic factors in shaping climate concern.

Additional data sources have further expanded the landscape of global climate opinion research. These include the World Risk Poll conducted by the Lloyd's Register Foundation (with waves in 2019, 2021, and 2023), which has reached 142 countries; the United Nations Development Programme's People's Climate Vote surveys (2021, 2024), which have covered 77 countries; and the Global Climate Change Survey (2022), which includes data from 125 countries.

Despite this progress, cross-national studies continue to face significant methodological challenges. Achieving comparable sampling across diverse countries remains difficult, and some regions–such as small island developing states or nations affected by conflict–are often underrepresented. Moreover, variation in question wording across surveys can complicate direct comparisons. The depth of survey instruments is another limitation; many global studies, including Gallup's, rely on only one or two items per country (e.g., binary yes/no questions on awareness and risk perception), limiting the ability to capture the full complexity of climate attitudes.

## 2.2 Longitudinal surveys

While recent years have seen a notable expansion in global coverage, much less progress has been made in building consistent time series that allow for longitudinal analysis of public attitudes. Most long-term panel or trend studies are confined to fewer than 20 countries, often centered on Europe and North America. For example, the Eurobarometer and the European Social Survey regularly include repeated items on climate change, but these datasets are largely restricted to EU or other European nations. Similarly, polling organizations in North America–particularly in the United States–have produced decades-long time series on climate beliefs and concern. However, comparable long-term datasets remain scarce outside the industrialized world (see Capstick et al. (2015) for a comprehensive review).

Among the few cross-national longitudinal initiatives that do exist, country diversity is often limited. The International Social Survey Programme (ISSP), for instance, has included environmental modules in 1993, 2000, 2010, and 2020, but each wave covered only around 20–30 countries, primarily higher-income nations in Europe, North America, and East Asia. In other world regions, regional barometer surveys have only recently begun to incorporate climate-related questions—often from 2019 onward. These include the Central Asia Barometer, the Arab Barometer, and the Afrobarometer. In the Americas, questions on climate change were introduced earlier—around 2010—via the Americas Barometer and Latinobarómetro. However, these surveys have not consistently repeated climate-related questions across time, limiting their usefulness for trend analysis.

In sum, longitudinal research on global public opinion toward climate change remains fragmented. Most long-term datasets disproportionately represent OECD countries, leaving significant knowledge gaps regarding trends in climate perception across much of Asia, Africa, and Latin America. This lack of continuous, comparable data from a broader set of countries hampers researchers' ability to determine whether global public opinion is converging or diverging along regional lines. It also constrains efforts to evaluate the long-term effectiveness of international climate communication and policy engagement strategies.

## 2.3 Conceptual heterogeneity and measurement dimensions

Finally, a persistent challenge in studying global climate opinion is the conceptual heterogeneity of available survey items. Instruments differ widely in what they measure—from awareness and risk perception to policy preferences and individual actions—making it difficult to ensure cross-national and longitudinal comparability. Drawing on insights from climate psychology and public opinion research, we identify five core dimensions commonly used to capture climate attitudes: awareness, human attribution, risk perception, policy support, and individual action. These dimensions, while interrelated, capture distinct aspects of public concern and serve as a framework for harmonizing diverse survey items. A more detailed description, along with examples, is provided in Appendix A.

# 3 Data and methods

To construct a continuous, country-year panel dataset of climate change support with near-global coverage, we first integrated data from multiple cross-national surveys into a unified dataset. This combined dataset includes responses from 3,915,823 individuals across 171 countries over a 15-year period (2010-2024), covering 76 distinct survey items.

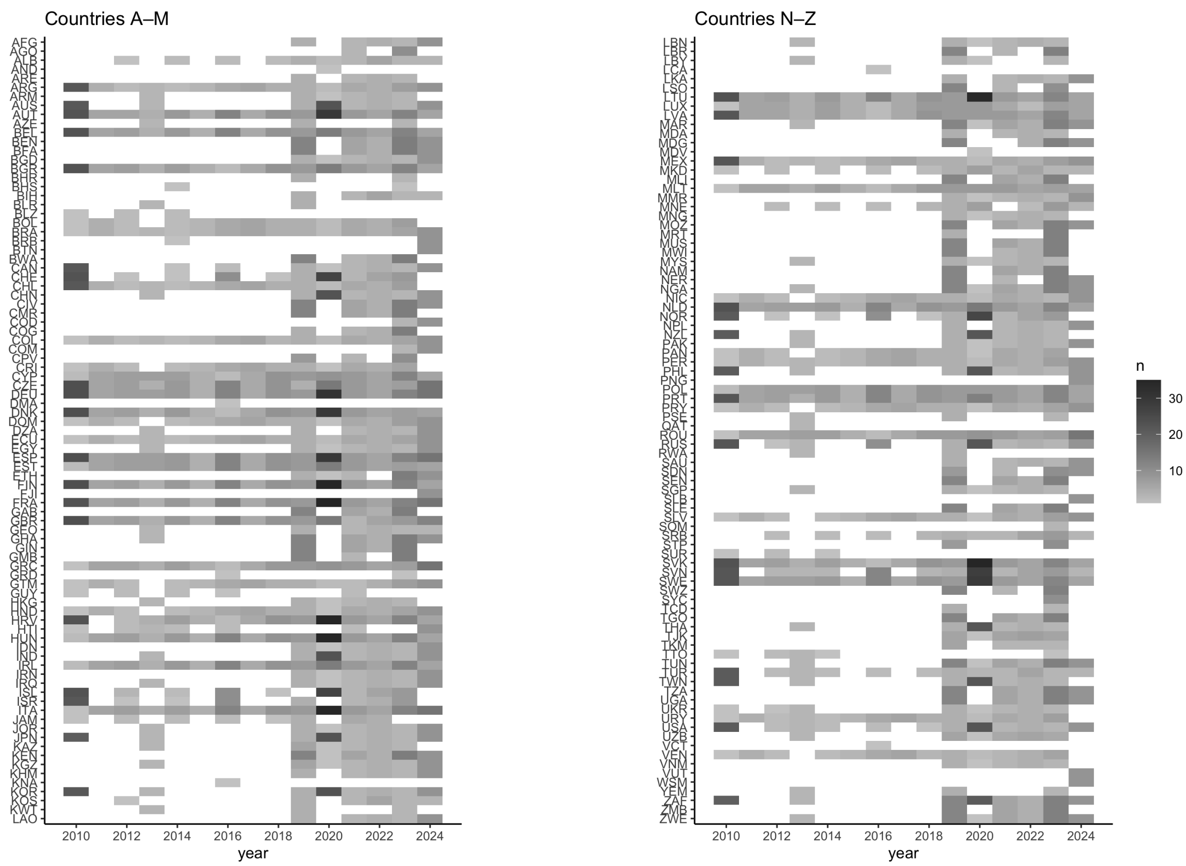
Measuring public attitudes across countries and over time presents significant methodological challenges. Survey data are often fragmented and heterogeneous, with substantial variation in question wording, response formats, sampling strategies, and geographic coverage—-complicating efforts at direct comparison. To address these issues, this section offers a descriptive overview of the data, focusing on three main aspects: the distribution of observations across countries and years (section 3.1; the attitudinal dimensions captured by the survey items (section 3.2); and the coherence among items, which is used to identify and exclude those that do not empirically align with the broader construct of climate change support (section 3.3). We close the section by introducing the empirical strategy (Bayesian Ordinal IRT) to build the new cross-country and longitudinal dataset (section 3.4).

## 3.1 Mapping the survey questions

We begin by systematically identifying and harmonizing all available cross-national surveys that include questions on climate change. Our dataset combines a broad set of global and regional sources conducted between 2010 and 2024, including the International Social Survey Programme (ISSP), Lloyd's Register Foundation's World Risk Poll, the Global Climate Change Survey, and regional barometers such as Latinobarómetro and Afrobarometer. For each survey, we document the year(s) of data collection, the number and geographic distribution of countries, the exact wording and structure of climate-related questions, the number of ordinal response categories, and the attitudinal dimension targeted by each item.[1]

[1] A detailed overview of included surveys, their coverage, and item characteristics is provided in Appendix B.

The resulting integrated dataset consists of 8,144 country-year–item observations, drawn from 12 major survey programs and encompassing 167 distinct climate-related questions. Figure 1 displays the temporal and geographic distribution of these observations. Each tile represents a country-year pair between 2010 and 2024; color intensity reflects the number of questions available for that combination—darker shades indicate more items, while lighter ones reflect fewer observations. Empty tiles correspond to country-years with no available data.




(a) Countries A–M

(b) Countries N–Z

Figure 1: Distribution of climate attitude items by country and year, split into two panels.

Figure 2 complements this by offering a global overview of survey coverage during the period 2010–2024. Countries shaded in darker tones exhibit higher data availability, usually due to the inclusion of multiple survey waves or items. Countries in white are excluded from the dataset due to a lack of available data.

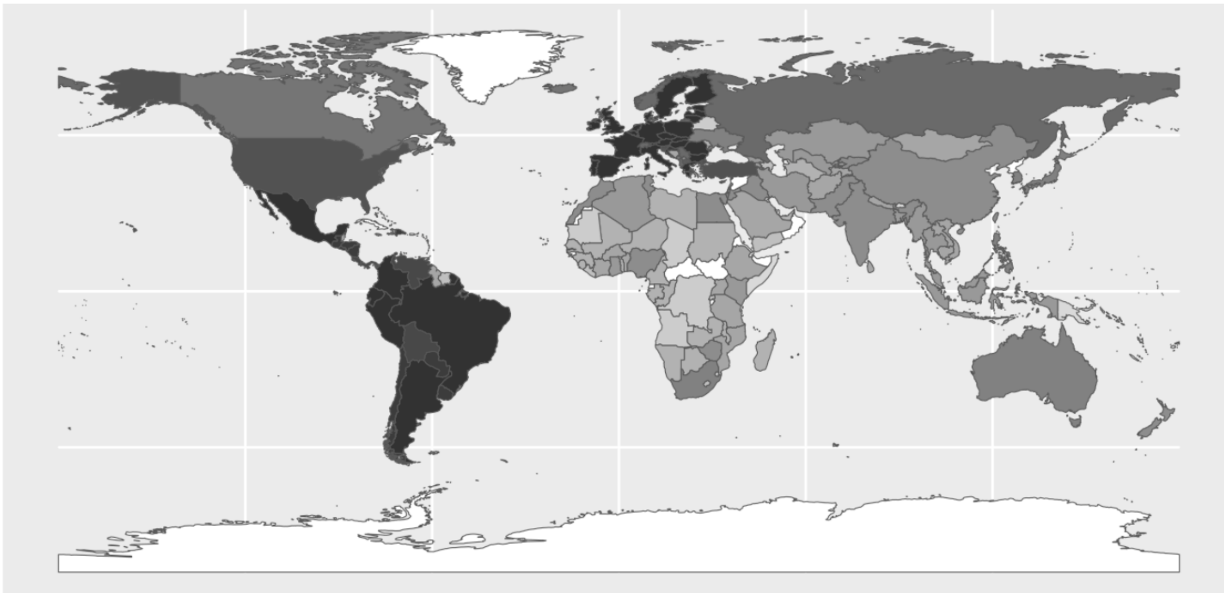


Figure 2: Global coverage of climate change attitude survey data (2010–2024).

Although global in ambition, coverage remains highly uneven. European countries benefit from frequent and extensive data collection through instruments such as the Eurobarometer and World Values Survey (WVS). In contrast, many countries in Sub-Saharan Africa and parts of Asia rely on less frequent, region-specific initiatives like Afrobarometer or Arab Barometer. Several countries—such as the Central African Republic, Cuba, North Korea, Eritrea, and Suriname—have no data at all. Small island nations and microstates (e.g., Kiribati, Tuvalu, Liechtenstein) also remain largely uncovered.

## 3.2 Exploring the empirical dimensions

To examine the underlying structure of public attitudes toward climate change, we apply Principal Component Analysis (PCA) to country-level mean scores of survey items. This exploratory technique allows us to detect clusters of questions that vary together across countries and over time. Our goal is to assess whether the most widely available items tap into a single latent dimension, or whether distinct empirical dimensions emerge.

We begin by standardizing each survey item to ensure cross-country comparability. Raw response counts are converted into proportions and rescaled into a normalized score ranging from 0 to 1. We retain only items with responses from at least 40 countries to ensure sufficient global coverage and estimation stability. This yields a subset of 27 climate-related items drawn from multiple sources.

To determine how many components to retain, we conduct a parallel analysis that compares observed eigenvalues to those derived from randomly simulated data. The resulting scree plot (Figure 3) shows a steep drop after the first component, with four components exceeding the simulation threshold. This pattern suggests the presence of a strong general factor, as well as evidence of additional underlying dimensions.

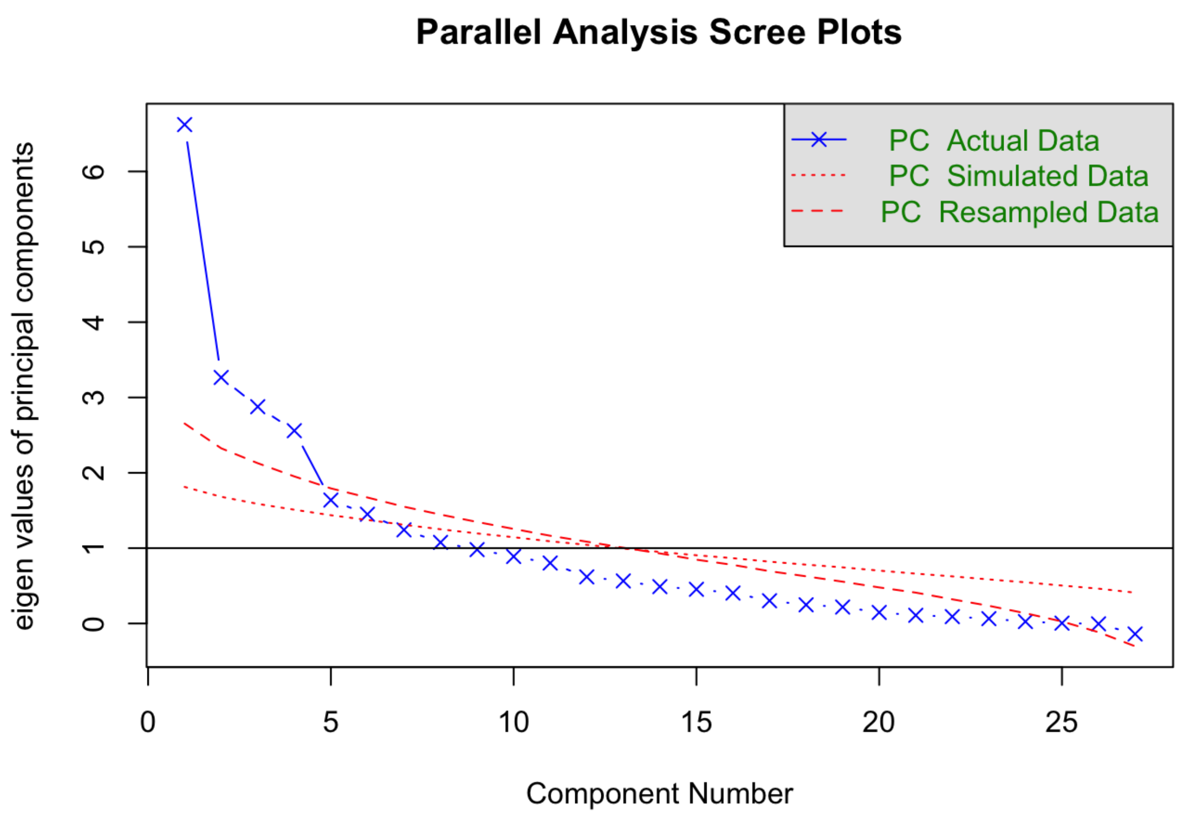



Figure 3: Parallel analysis scree plot for PCA.

The first component explains a substantial share of variance and loads positively across a broad range of items from major global datasets such as the World Risk Poll, People's

Climate Vote, and the Global Climate Change Survey. As illustrated in Figure 4, this component likely reflects a general latent construct of climate concern and support that cuts across item wording, response scales, and survey instruments.

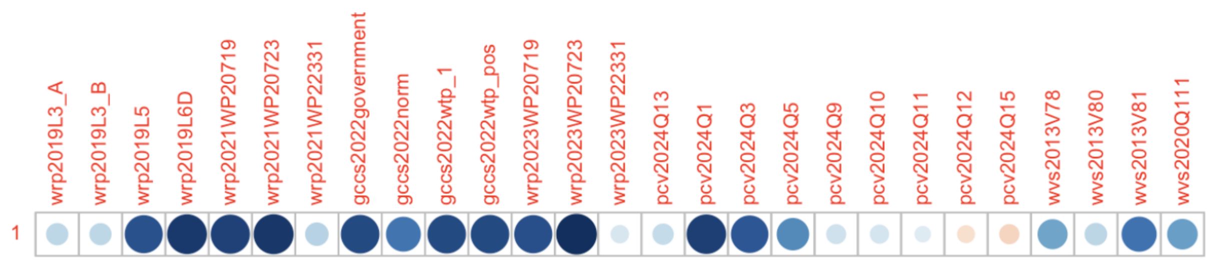



Figure 4: Loadings on first principal component.

In sum, these results provide empirical support for a unidimensional structure, suggesting that a broad range of survey items can be meaningfully integrated into a single climate attitude index.[2]

### 3.3 Item screening and selection

Building on the previous analysis, we now assess the empirical coherence of lower-coverage survey items—primarily from regional barometers and irregularly fielded global surveys—to determine their alignment with the core latent structure of climate attitudes. The goal is to retain only those items that meaningfully contribute to a unified, cross-national measurement model, while avoiding the inclusion of indicators that introduce measurement noise or conceptual divergence.

We begin by identifying the most representative items from high-coverage surveys based on their loadings on the first principal component (Figure 4). Using a loading threshold of 0.6, we retain 11 items that consistently align with the core latent dimension of climate

[2]While a unidimensional model is appropriate for the main analysis, the parallel analysis also supports the extraction of up to four components. Each represents a distinct thematic cluster of items. See Appendix C for further discussion and loading plots.

concern and support. These items, summarized in Table 1, serve as empirical anchors for evaluating the coherence of additional survey questions.

Table 1: Core climate attitude survey items

| Dataset | Year | Item code |
|---|---|---|
| World Risk Poll | 2019 | L5 |
| World Risk Poll | 2019 | L6D |
| World Risk Poll | 2021 | WP20719 |
| World Risk Poll | 2021 | WP20723 |
| Global Climate Change Survey | 2022 | government |
| Global Climate Change Survey | 2022 | wtp_1 |
| Global Climate Change Survey | 2022 | wtp_pos |
| World Risk Poll | 2023 | WP20719 |
| World Risk Poll | 2023 | WP20723 |
| People's Climate Vote | 2024 | Q1 |
| People's Climate Vote | 2024 | Q3 |

To assess the alignment of additional items, we compute their correlation with a proxy index derived as the row-wise mean of these 11 anchor items. This simple composite avoids the instability of PCA-based scores in sparse datasets and offers an interpretable benchmark for coherence.[3]

Figure 5 summarizes the distribution of correlations between each candidate item and the core index. Most items from larger surveys (15+ countries) fall within the moderate correlation range of 0.3 to 0.6, indicating acceptable conceptual alignment. Notably, the highest correlations are found among items from low-coverage surveys, which raises no redundancy concerns due to their limited geographic scope. Only four items show negative correlations, all of which come from low-coverage sources, suggesting some potential misalignment but with minimal impact on overall scale integrity.

Following this screening process, we retain a refined set of 102 item-survey-year observa-

[3]We apply two criteria to determine item retention. First, a *correlation threshold*: items must correlate at least 0.3 with the anchor index. This relatively inclusive cutoff balances the need for consistency with tolerance for contextual variation in question wording, time, or region. Second, a *coverage threshold*: items asked in fewer than 15 countries are retained regardless of their correlation to preserve regional diversity and representation. By contrast, widely used items (15+ countries) that fail to meet the correlation criterion are excluded.

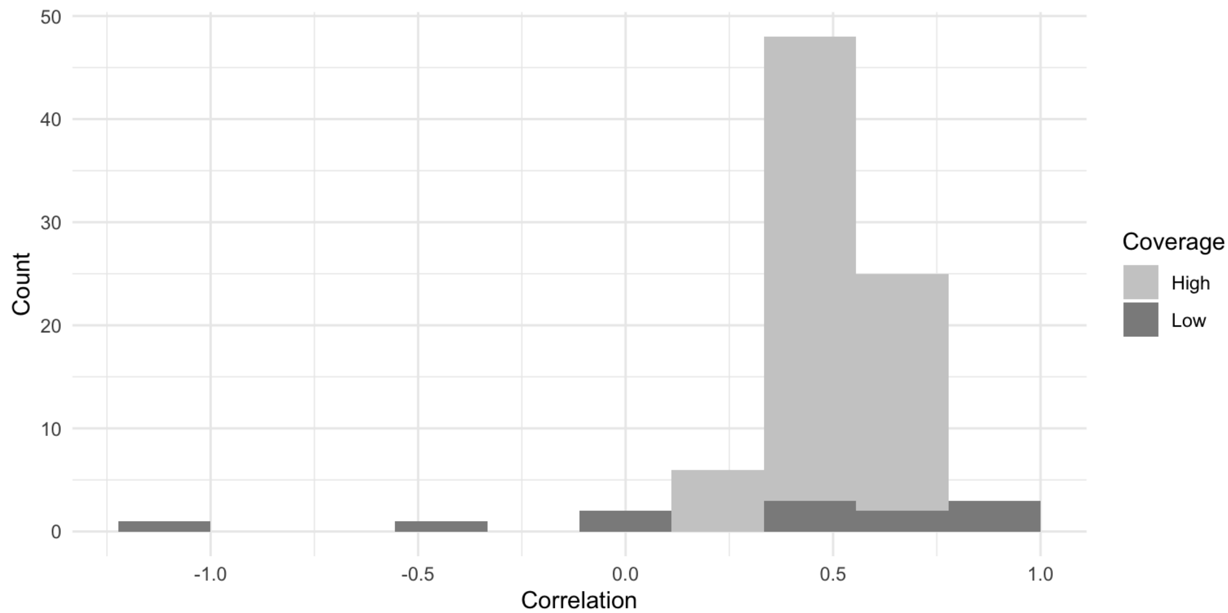



Figure 5: Correlations between retained survey items and the anchor index.

tions that meet either the correlation or coverage thresholds. These items span 76 unique question formulations, cover 171 countries, and include responses from over 3.9 million individuals between 2010 and 2024. This final dataset balances empirical coherence with inclusiveness—preserving conceptual consistency while maintaining broad geographic and thematic coverage. It provides a solid foundation for estimating a latent climate support scale that captures global variation in public attitudes with improved validity and reach.

### 3.4 Estimating climate attitudes using Bayesian Ordinal IRT

Our empirical estimation model will be implemented using Bayesian multilevel estimation, which pools information across countries, years, and survey items. This approach effectively addresses data sparsity and enables the integration of heterogeneous data sources.

As noted earlier, data coverage remains highly uneven: European countries benefit from frequent and rich data collection, while many countries in Sub-Saharan Africa and parts of Asia are less consistently covered or even entirely missing. This spatial and temporal unevenness poses significant challenges for producing complete and reliable country-year estimates. In data-rich contexts, the model leverages multiple overlapping observations, whereas in sparse contexts, gaps are mitigated through hierarchical modeling and partial pooling.

To accommodate these issues, we treat each survey item as an observed indicator of one or more underlying latent variables. By applying Bayesian ordinal item response models, we integrate heterogeneous, noisy, and incomplete survey data within a coherent latent variable framework. This enables the estimation of continuous, country-year-level measures of public climate change attitudes that explicitly account for uncertainty and measurement error[4].

Through partial pooling across countries, years, and items, the model effectively leverages information across groups and mitigates issues arising from uneven coverage. The resulting framework is flexible and robust, generating comparable latent attitude scores across countries and over time.[5]

Beyond this static model, we extend the specification to incorporate temporal smoothing by treating climate attitudes as a dynamic latent trait evolving over time. This imposes structure on the year dimension, enabling estimation even for country-years with missing data. The dynamic extension preserves meaningful longitudinal trends while reducing noise arising from sparse or irregular survey distributions. The outcome is a temporally smoothed, globally harmonized panel of latent climate attitudes.

# 4 Expected results

This section outlines the next analytical steps and presents the structure of the expected results. At this stage, we do not yet report empirical estimates or model outputs. Rather, we describe three complementary strategies that we plan to implement to generate a comparative dataset on climate change attitudes.

[4]A detailed discussion of handling ordinal variables is provided in Appendix D, and information on data formatting and response types can be found in Appendix E.

[5]The model is estimated within a Bayesian framework using the `brms` package in R, with the `cmdstanr` backend for efficient sampling. We employ the cumulative logit link function, consistent with the graded response model (GRM) specification, running 600 iterations per chain (300 warm-up) across four chains.

All three strategies are built upon the unidimensional latent structure identified in the previous section and rely on a shared modeling framework: a Bayesian ordinal Item Response Theory (IRT) model with partial pooling. While conceptually consistent, each strategy prioritizes different trade-offs between geographic breadth, temporal depth, and data density. As such, the resulting estimates for a given country-year may differ slightly across approaches.

Specifically, we plan to produce:

1. A cross-sectional dataset with maximum country coverage for the most recent year with sufficient data availability;

2. A balanced country-year panel spanning the 2010–2024 period for countries with consistent coverage;

3. A long-term time series extending back to the early 1990s, focusing on a more limited set of countries (primarily within the OECD) where historical data are available.

Each of these outputs is designed to serve different analytical goals–from mapping global attitudes at a single point in time to enabling time-series analysis of opinion dynamics. The subsections below describe the logic, expected coverage, and modeling implications of each strategy.

## 4.1 Cross-sectional country data

Building on the exploratory analysis from the previous section, we assess the maximum cross-national coverage achievable in a single-year dataset of climate change attitudes. Our results suggest that it is feasible to construct a cross-sectional dataset encompassing at least 167 countries, drawing on data collected between 2019 and 2024. This recent period marks a turning point in the availability of globally comparable climate opinion data, due to a rapid expansion in both geographic reach and survey frequency.

The launch of the World Risk Poll in 2019 initiated a new wave of large-scale, cross-national climate surveys.[6] It was followed by other major initiatives, including the People's Climate Vote (2021) and the Global Climate Change Survey (2022). At the same time, several regional barometers expanded their coverage of climate-related questions: Afrobarometer incorporated climate items starting in Round 7 (2019), and the Arab Barometer introduced them for the first time.

By aggregating data from these global and regional surveys–and allowing for limited borrowing from adjacent years–we will be able to assemble the most comprehensive cross-sectional dataset on global climate attitudes to date. Table 2 summarizes the regional distribution of countries with at least one valid observation in the 2019–2024 period.

Table 2: Countries per region with at least one observation (2019–2024)

| **Region** | **Countries** | **Total** | **Coverage (%)** |
|---|---|---|---|
| North America | 2 | 2 | 100.0 |
| South Asia | 8 | 8 | 100.0 |
| Europe & Central Asia | 50 | 53 | 94.3 |
| Sub–Saharan Africa | 42 | 48 | 87.5 |
| Middle East & North Africa | 17 | 21 | 81.0 |
| Latin America & Caribbean | 25 | 33 | 75.8 |
| East Asia & Pacific | 22 | 32 | 68.8 |

These 167 countries–out of the 198 included in the ISO3c classification–account for 99.5% of global GDP and 98.6% of the world's population. Politically, the sample is reasonably balanced: 62% of included countries are classified as democracies, compared to 58.3% among those not covered. However, economic disparities persist: the average GDP per capita of included countries is 24,060€, versus 18,448€ for excluded ones.

## 4.2 Country-year panel dataset

Beyond enabling broad cross-sectional analysis, the data also support the construction of a longitudinal country-year panel–an important resource for tracking the evolution of public

[6]The Gallup World Poll 2010 remains an early exception.

support for climate action over time. Drawing on surveys conducted between 2010 and 2024, we identify 72 countries with repeated climate-related observations, allowing for the creation of a balanced panel dataset spanning 15 years.

However, this temporal coverage remains uneven across regions (see Figure 1). Table 3 summarizes the number of countries within each world region that have at least two valid survey waves between 2010 and 2018. While North America is fully covered, and coverage is relatively strong in Latin America and Europe, it is much more limited in other regions—particularly Sub-Saharan Africa and South Asia.

Table 3: Countries per region with ≥2 observations between 2010–2018

| **Region** | **Countries** | **Total** | **Coverage (%)** |
|---|---|---|---|
| North America | 2 | 2 | 100.0 |
| Latin America & Caribbean | 24 | 33 | 72.7 |
| Europe & Central Asia | 37 | 53 | 69.8 |
| East Asia & Pacific | 6 | 32 | 18.8 |
| Middle East & North Africa | 2 | 21 | 9.5 |
| Sub-Saharan Africa | 1 | 48 | 2.1 |
| South Asia | 0 | 8 | 0.0 |

While limited in some regions, this panel offers valuable leverage for studying climate opinion dynamics in middle- and high-income democracies, particularly in Latin America and Europe.[7]

## 4.3 Longitudinal dataset

A final strategy prioritises temporal depth over cross-sectional breadth by extending the dataset backward in time. Under this approach, we estimate that a longitudinal panel could be constructed for approximately 50 countries, reaching as far back as the early 1990s. This historical extension would rely primarily on European countries, where repeated observations

[7]Future expansions are also possible. Incorporating the 2010 wave of the Gallup World Poll would extend the panel to nearly 100 countries. In addition, older surveys—such as Wave 5 of the World Values Survey (2005–2009)—may allow for historical comparisons in selected cases, especially within the OECD. These additions would support longer time series and enable deeper regional analysis of long-term attitudinal trends.

are available through multiple waves of the Eurobarometer and the European Social Survey. In addition, a broader set of OECD countries can be included using earlier waves of the World Values Survey and the International Social Survey Programme (notably from 1993 and 2000), as well as select waves of Latinobarómetro. While such coverage is necessarily limited to countries with long-standing survey infrastructure, it offers valuable opportunities to analyse long-term attitudinal change in climate attitudes.

# 5 Conclusions

The aim of this research is to present a harmonized, globally representative dataset of public attitudes toward climate change, covering more than 150 countries over the period 2010–2024. By integrating diverse survey instruments and applying a Bayesian ordinal IRT model, we aim at generating a latent measure of climate support that will enable drawing consistent cross-national and longitudinal comparison. The dataset will substantially improve temporal continuity and geographic inclusiveness, particularly in underrepresented regions. It will also offers a robust empirical foundation for studying global opinion dynamics on climate change and supports future research on the political, economic, and institutional drivers of public climate concern.

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# A Conceptual Dimensions of Climate Attitudes

The measurement of public attitudes toward climate change is often fragmented and inconsistent. Surveys differ in the specific aspects of climate opinion they target, including awareness, perceived risks, causal beliefs, policy preferences, and individual behaviours. This diversity complicates efforts to compare results across countries or time, as survey items may not be directly equivalent.

To systematize this variation, we draw on existing literature in climate psychology and public opinion research (Leiserowitz, 2005; Lee et al., 2015; Hornsey et al., 2016; Poortinga et al., 2019) to identify five commonly used and theoretically grounded dimensions of climate attitudes:

1. **Awareness**: Whether individuals have heard of climate change and believe it is occurring.

2. **Human Attribution**: The belief that climate change is primarily caused by human activities.

3. **Risk Perception**: The extent to which climate change is seen as a serious personal, national, or global threat.

4. **Policy Support**: Willingness to endorse governmental responses, such as emissions regulations or renewable subsidies.

5. **Individual Action**: Self-reported behaviours or intentions aimed at mitigating climate change, including acceptance of personal costs.

These dimensions are often positively correlated: for example, awareness tends to predict greater concern, which in turn relates to stronger attribution of human causes and higher levels of support for mitigation policies (Lee et al., 2015; Poortinga et al., 2019; Mayer et al., 2017). While other constructs—such as institutional trust or perceived policy efficacy—may also be relevant in some contexts (Smith and Mayer, 2018; Drews and van den Bergh, 2016), the five dimensions identified here offer a parsimonious and widely applicable framework.

Table A.1 provides definitions and representative examples for each dimension.

Table A.1: Dimensions of climate change

| **Dimension** | **Description** | **Example** |
|---|---|---|
| Awareness | Recognition that climate change exists or is occurring. | "Generally speaking, how concerned are you about environmental issues?" (International Social Survey Program) |
| Risk perception | Concern about the potential impacts of climate change. | "Do you think climate change will be a serious threat to your country in the next 20 years?" (Lloyd's Register Foundation) |
| Attribution to human causes | Belief that human activity is a primary driver of climate change. | "Please tell me whether you strongly agree, agree, disagree or strongly disagree with the following statements: Human beings are the main responsible for climate change?" (Latinobarometro) |
| Policy support | Willingness to endorse governmental policies aimed at reducing emissions or mitigating climate change. | "Do you think the national government should do more to fight global warming?" (Global Climate Change Survey) |
| Personal responsibility | Self-reported behavioral change or intention to act at the individual level. | "How often do you make a special effort to sort glass or tins or plastic or newspapers and so on for recycling?" (International Social Survey Program) |

## B Surveys and questions

Table B.1 provides a summary of the surveys used, indicating each source's release year and number of climate-related items. These include both one-off cross-sectional surveys and repeated cross-wave surveys that enable tracking over time. Each question was classified into one of the five attitudinal dimensions described above.

Table B.1: Global and regional surveys measuring climate change attitudes

| Survey | Year | Items |
|---|---|---|
| Afrobarometer R7 | 2019 | 8 |
| Afrobarometer R8 | 2021 | 3 |
| Afrobarometer R9 | 2023 | 10 |
| Americas Barometer 2010 | 2010 | 1 |
| Americas Barometer 2012 | 2012 | 2 |
| Americas Barometer 2014 | 2014 | 2 |
| Americas Barometer 2017 | 2017 | 2 |
| Americas Barometer 2019 | 2019 | 1 |
| Americas Barometer 2021 | 2021 | 2 |
| Americas Barometer 2023 | 2023 | 1 |
| Arab Barometer V | 2019 | 3 |
| Arab Barometer VI | 2021 | 1 |
| Arab Barometer VII | 2022 | 6 |
| Arab Barometer VIII | 2024 | 21 |
| Central Asia Barometer Wave 5 | 2019 | 2 |
| Central Asia Barometer Wave 10 | 2021 | 4 |
| Central Asia Barometer Wave 11 | 2022 | 4 |
| Central Asia Barometer Wave 12 | 2023 | 4 |
| European Social Survey Round 5 | 2010 | 1 |
| European Social Survey Round 6 | 2012 | 1 |
| European Social Survey Round 7 | 2014 | 1 |
| European Social Survey Round 8 | 2016 | 9 |
| European Social Survey Round 9 | 2018 | 1 |
| European Social Survey Round 10 | 2020 | 4 |
| European Social Survey Round 11 | 2023 | 4 |
| Eurobarometer Special: Climate change | 2012 | 4 |
| Eurobarometer Special: Climate change | 2013 | 4 |
| Eurobarometer Special: Climate change | 2015 | 4 |
| Eurobarometer Special: Climate change | 2017 | 5 |
| Eurobarometer Special: Climate change | 2019 | 4 |
| Eurobarometer Special: Climate change | 2021 | 5 |
| Eurobarometer Special: Climate change | 2023 | 4 |
| Eurobarometer Special: Attitudes of Europeans towards the Environment | 2011 | 6 |

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Table B.1: Global and regional surveys measuring climate change attitudes (continued)

| Survey | Year | Items |
|---|---|---|
| Eurobarometer Special: Attitudes of Europeans towards the Environment | 2014 | 6 |
| Eurobarometer Special: Attitudes of Europeans towards the Environment | 2018 | 6 |
| Eurobarometer Special: Attitudes of Europeans towards the Environment | 2020 | 6 |
| Eurobarometer Special: Attitudes of Europeans towards the Environment | 2024 | 3 |
| Eurobarometer Standard | 2010 | 3 |
| Eurobarometer Standard | 2012 | 2 |
| Eurobarometer Standard | 2014 | 2 |
| Eurobarometer Standard | 2016 | 2 |
| Eurobarometer Standard | 2018 | 2 |
| Eurobarometer Standard | 2020 | 2 |
| Eurobarometer Standard | 2022 | 2 |
| Eurobarometer Standard | 2024 | 3 |
| Global Climate Change Survey - Gallup World Poll 2021/2022 | 2022 | 4 |
| International Social Survey Program | 2010 | 21 |
| International Social Survey Program | 2020 | 21 |
| Latinobarometro 2011 | 2011 | 4 |
| Latinobarometro 2015 | 2015 | 3 |
| Latinobarometro 2016 | 2016 | 3 |
| Latinobarometro 2017 | 2017 | 6 |
| Latinobarometro 2018 | 2018 | 3 |
| Latinobarometro 2020 | 2020 | 1 |
| Latinobarometro 2023 | 2023 | 3 |
| People's Climate Vote | 2024 | 9 |
| World Risk Poll - Lloyd's Register Foundation | 2019 | 4 |
| World Risk Poll - Lloyd's Register Foundation | 2021 | 3 |
| World Risk Poll - Lloyd's Register Foundation | 2023 | 3 |
| World Values Survey 6 | 2016 | 3 |
| World Values Survey 7 | 2023 | 1 |

As a first tentative categorization, each item was assigned to one of five core dimensions of climate change attitudes through a combination of manual coding and AI-assisted classification. The resulting distribution of items is as follows: the dimension with the highest number of items is risk perception (93 items), closely followed by policy support (78 items). The remaining dimensions are awareness (43 items), personal responsibility (39 items), and human attribution (25 items). Figure B.1 illustrates the distribution of survey items across these five dimensions.

Notably, this distribution of items across dimensions closely mirrors the empirical patterns identified in subsequent analyses. This alignment between the initial theoretical framework and the observed empirical structure suggests that the core dimensions—risk perception, policy support, awareness, personal responsibility, and human attribution—capture meaningful and distinct facets of public attitudes toward climate change across countries

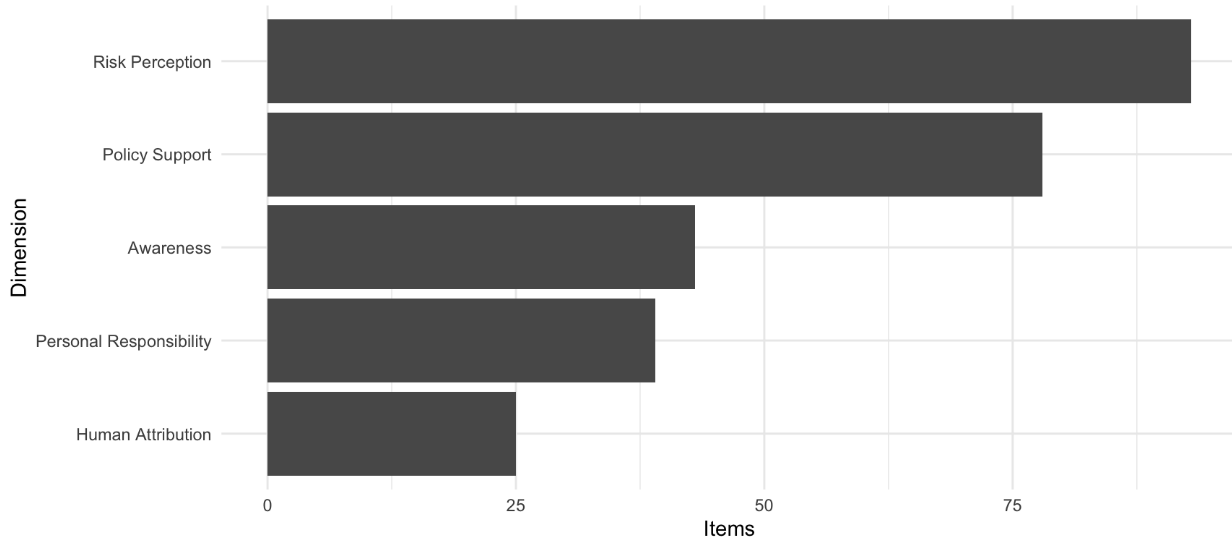


Figure B.1: Distribution of survey items across the five dimensions of climate change attitudes.

and surveys.

# C Principal Component Structure

To explore the underlying structure of the climate attitude items, we conducted a principal component analysis (PCA). While a single dominant factor is evident, the results of a parallel analysis suggest that up to four components may be meaningfully extracted. The loadings shown in Figure C.1 illustrate how each additional component captures distinct thematic clusters of survey items:

- **TC3 – Climate Threat Perception**: Items loading strongly on this component refer to national vulnerability, fear of severe weather, and concern for future generations.
- **TC1 – Personal Responsibility**: This component captures items emphasizing individual engagement and willingness to accept personal costs or behavioral change.
- **TC2 – Localized Environmental Threat**: Reflects awareness of immediate, personally salient environmental risks (e.g., effects on daily life or safety).
- **TC4 – Policy Support**: Items here focus on collective and institutional action to address climate change, such as government regulation or international agreements.

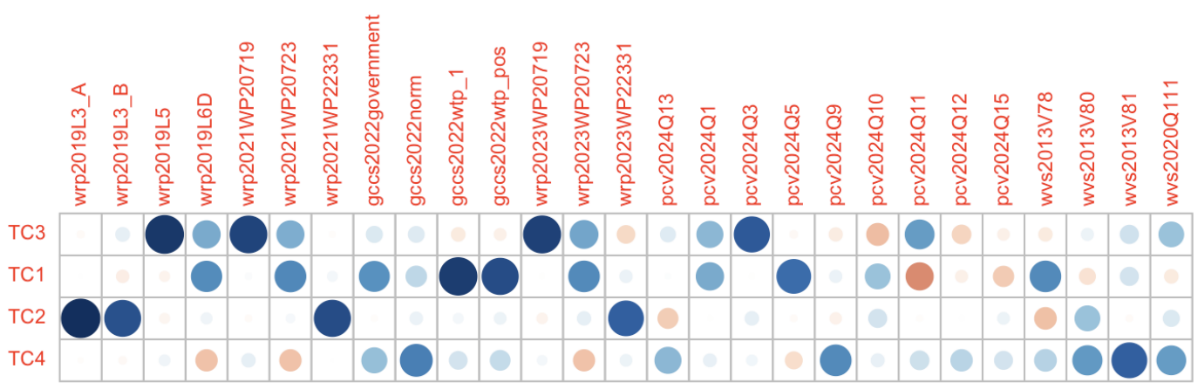


Figure C.1: Loadings on the first four principal components of climate attitude items. Each component reflects a distinct thematic dimension beyond the general support factor.

# D Handling ordinal data

A key challenge in cross-national opinion research is managing diverse response formats: binary (e.g., "yes"/"no"), ordinal (e.g., "strongly agree," "agree," "neutral"), and multi-category scales (which can be treated as either ordinal or interval data). These variations complicate comparisons of public attitudes toward climate change across different surveys. Binary responses are relatively straightforward to handle using methods such as mean calculations, logistic regression, or dichotomous IRT models.

However, non-binary responses, particularly ordinal data, are more complex. More response categories allow for finer-grained information, but they also make aggregation and modeling more difficult. One option is to convert ordinal categories into numeric values (e.g., scoring responses from 0 to 1). This produces normalized variables that simplify modeling and comparisons. However, this method assumes equal spacing between categories—a problematic assumption. For example, a score of 2.5 on a four-point scale is not equivalent to 2.5 on a five-point scale. Additionally, this approach loses detail on the distribution of responses across categories.

Another strategy involves creating one variable per response category. In this method, if a question has three possible responses, three variables are created in the dataset, each representing the count of respondents in that category, along with a fourth variable representing the total number of respondents. This structure preserves the full distribution of responses and allows for flexible post-processing. However, it results in a wider dataset and raises the additional complexity of tracking missing values per category.

Despite the added complexity, we adopt the disaggregated count approach. This method retains the ordinal nature of the data and is well-suited for models that treat category responses probabilistically. Specifically, we apply a Bayesian Item Response Theory (IRT) model, where the observed distribution across response categories is modeled as a function of a latent attitudinal trait. The model estimates the probability of each response, conditional on the underlying level of public support.

In this format, the resulting dataset is organized with the country-year-question as the unit of observation. Table D.1 illustrates the first rows of the dataset.

Table D.1: Distribution of survey responses by country-year-item

| country | year | db | cdb | c1 | c2 | c3 | c4 | cats | n_na | total |
|---|---|---|---|---|---|---|---|---|---|---|
| Afghanistan | 2019 | wrp | L3_A | 23 | 1097 | NA | NA | 2 | 7 | 1127 |
| Afghanistan | 2019 | wrp | L3_B | 49 | 1073 | NA | NA | 2 | 5 | 1127 |
| Afghanistan | 2019 | wrp | L5 | 415 | 452 | 102 | NA | 3 | 158 | 1127 |
| Afghanistan | 2019 | wrp | L6D | 401 | 521 | 198 | NA | 3 | 7 | 1127 |
| Afghanistan | 2021 | wrp | WP20719 | 366 | 447 | 101 | NA | 3 | 86 | 1000 |
| Afghanistan | 2021 | wrp | WP20723 | 305 | 461 | 209 | NA | 3 | 25 | 1000 |
| Afghanistan | 2021 | wrp | WP22331 | 5 | 962 | NA | NA | 2 | 33 | 1000 |
| Afghanistan | 2022 | gccs | government | 844 | 118 | NA | NA | 2 | 38 | 1000 |
| Afghanistan | 2022 | gccs | norm | 852 | 106 | NA | NA | 2 | 42 | 1000 |

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Table D.1: Distribution of survey responses by country-year-item (continued)

| country | year | db | cdb | c1 | c2 | c3 | c4 | cats | n_na | total |
|---|---|---|---|---|---|---|---|---|---|---|
| Afghanistan | 2022 | gccs | wtp_1 | 818 | 156 | NA | NA | 2 | 26 | 1000 |

Each response category is stored as a separate column (c1 to c4), where c1 always represents the highest level of agreement, concern, or support (e.g., "Very serious threat"); higher-numbered columns (e.g., c2, c3, c4) correspond to lower levels on the scale; cats records the number of response categories for that question; n_na captures the number of missing responses, and total indicates the total number of respondents for that item.

For binary questions, only c1 and c2 are filled; the others remain NA. For ordinal questions with three or four categories, the respective number of c columns are populated accordingly. This consistent structure allows us to preserve ordinal detail while enabling robust modeling of public attitudes.

# E Building the data for PCA

To explore the empirical structure of the data, we begin by constructing a standardized, comparable measure for each survey item. Our dataset captures, for every country-year-dataset-item observation, the number of respondents selecting each response category. However, these items differ in the number of response categories—some are binary, while others are multi-point ordinal scales ranging up to 11 categories.

To harmonize these variations, we first convert the raw counts into proportions by dividing each response count by the number of valid (non-missing) responses. We then compute a mean response score for each observation, designed to reflect the overall level of agreement or concern expressed by the public. This score is calculated using a weighted scheme that assumes higher response categories indicate stronger support or concern. Following standard practice in latent trait modeling, we assume that ordinal responses represent increasing levels along an underlying continuum, although the exact distances between categories are not treated as fixed (Samejima, 1969; Treier and Jackman, 2008).

This approach yields a normalized mean score bounded between 0 and 1, allowing for consistent interpretation across items with differing numbers of categories. The logic of this procedure aligns with prior efforts to recover latent public attitudes from heterogeneous survey instruments, particularly in the domain of democratic support (Claassen, 2019; Lupu and Zechmeister, 2021).

The output is a wide-format dataset consisting of 173 rows, each representing a country, and 232 columns, each representing a survey item. Each cell contains the average normalized response for a specific country-year-item combination. This structure, illustrated in Table E.1, facilitates the analysis of inter-item correlations and helps assess whether different survey questions reflect a common underlying dimension of climate change attitudes.

Table E.1: Wide-format dataset with continuous data

| Country | wrp2019L3_A | wrp2019L3_B | wrp2019L5 | wrp2019L6D | wrp2021WP20719 |
|---|---|---|---|---|---|
| Afghanistan | 0.021 | 0.044 | 0.662 | 0.591 | 0.645 |
| Albania | 0.012 | 0.013 | 0.747 | 0.526 | 0.794 |
| Algeria | 0.035 | 0.027 | 0.583 | 0.487 | 0.653 |
| Andorra | NA | NA | NA | NA | NA |
| Angola | NA | NA | NA | NA | NA |
| Argentina | 0.002 | 0.008 | 0.851 | 0.646 | 0.835 |
| Armenia | 0.031 | 0.028 | 0.631 | 0.348 | 0.677 |
| Australia | 0.030 | 0.027 | 0.704 | 0.403 | 0.723 |
| Austria | 0.029 | 0.027 | 0.792 | 0.554 | 0.789 |
| Azerbaijan | 0.061 | 0.057 | 0.629 | 0.378 | NA |

While different surveys ask different questions, the key issue is whether these variations significantly alter the relative rankings of countries. If a country that ranks high in climate change awareness also ranks high in perceived threat and policy support, then treating different dimensions as interchangeable becomes less problematic.